

Seed 8, Layer 2: The bracket tower $f_k^{(m)}$, exact q -monotonicity, and the Divisibility Transfer Lemma

Seed 8 / Layer 2 / Round 2

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1 Setup

Fix $r = 3$, level $d \not\equiv 0 \pmod{3}$, and a profile $c = (c_0, c_1, c_2)$ with $c_0 + c_1 + c_2 = d$. Let $g_m = g_m(q)$ denote the generating function for cylindric partitions of profile c with largest part exactly m (so $g_0 = 1$), computed by the profile-space transfer matrix. Set $h_m = (q; q)_m g_m$, and define the tower of Round 1:

$$D_0^m = h_m, \quad D_k^m = D_{k-1}^m - q^k D_{k-1}^{m-1} \quad (1 \leq k \leq m), \quad Q_n = D_n^n.$$

2 The bracket recursion

Define $f_{-1}^{(m)} := g_m$ and, for $k \geq 0$ and $m \geq k + 1$,

$$f_k^{(m)} := (1 - q^{m-k}) f_{k-1}^{(m)} - q^{k+1} f_{k-1}^{(m-1)}. \quad (1)$$

Lemma 1 (T1). *For all $k \geq -1$ and $m \geq k + 1$,*

$$D_{k+1}^m = (q; q)_{m-k-1} f_k^{(m)}.$$

Proof. Induction on k . Base $k = -1$: $D_0^m = h_m = (q; q)_m g_m = (q; q)_m f_{-1}^{(m)}$ by definition. Inductive step: assume the claim at $k - 1$ for all admissible m . Then

$$\begin{aligned} D_{k+1}^m &= D_k^m - q^{k+1} D_{k-1}^{m-1} = (q; q)_{m-k} f_{k-1}^{(m)} - q^{k+1} (q; q)_{m-1-k} f_{k-1}^{(m-1)} \\ &= (q; q)_{m-k-1} \left[(1 - q^{m-k}) f_{k-1}^{(m)} - q^{k+1} f_{k-1}^{(m-1)} \right] = (q; q)_{m-k-1} f_k^{(m)}, \end{aligned}$$

using $(q; q)_{m-k} = (q; q)_{m-k-1} (1 - q^{m-k})$ and recursion (1). □

Corollary 1 (T2). $Q_n = D_n^n = (q; q)_0 f_{n-1}^{(n)} = f_{n-1}^{(n)}$.

Hence Warnaar's positivity conjecture is exactly the diagonal $m = k + 1$ of the bracket tower.

Lemma 2 (Closed form). *For $k \geq -1$ and $m \geq k + 1$,*

$$f_k^{(m)} = \sum_{i=0}^{k+1} (-1)^i q^{\binom{i+1}{2}} \begin{bmatrix} k+1 \\ i \end{bmatrix}_q (q^{m-k}; q)_{k+1-i} g_{m-i}.$$

Proof. Combine Lemma 1 with the Round-1 closed form $D_{k+1}^m = \sum_i (-1)^i q^{\binom{i+1}{2}} \begin{bmatrix} k+1 \\ i \end{bmatrix}_q h_{m-i}$, divide by $(q; q)_{m-k-1}$, and use $h_{m-i} = (q; q)_{m-i} g_{m-i}$ together with $(q; q)_{m-i} / (q; q)_{m-k-1} = (q^{m-k}; q)_{k+1-i}$ for $0 \leq i \leq k + 1$. (Each quotient is a genuine polynomial, so the identity holds in $\mathbb{Z}[[q]]$.) □

3 The Master Conjecture (verified, not proved)

Conjecture 1 (Master). *For all $k \geq -1$, $m \geq k + 1$, and $0 \leq j \leq m - k - 1$,*

$$(q; q)_j f_k^{(m)} \succeq 0$$

(coefficientwise), and this is exact: $(q; q)_{m-k} f_k^{(m)}$ has a negative coefficient. In particular ($k = -1$): g_m is exactly m -fold q -monotone.

The top case $j = m - k - 1$ is $D_{k+1}^m \succeq 0$ (the Round-1 tower); the case $(k, j) = (m - 1, 0)$ is $Q_m \succeq 0$. Verified computationally, with exactness at every boundary, for: $d = 2$ ($m \leq 8, k \leq 4$); $d = 4$, all 5 orbit representatives ($m \leq 7, k \leq 4$); $d = 5$, all 7 representatives ($m \leq 6, k \leq 4$); $d = 7$, all 6 representatives ($m \leq 5, k \leq 3$).

Multiplying (1) by $(q; q)_j$ shows the induction on k needs, at every level, the full “monotonized domination” $(q; q)_j (1 - q^{m-k-1}) f_k^{(m)} \succeq q^{k+2} (q; q)_j f_k^{(m-1)}$; positivity alone does not chain. The one general reduction we can prove is:

Lemma 3 (Divisibility Transfer). *$\frac{1 - q^a}{1 - q^b}$ has nonnegative coefficients iff $b \mid a$. Consequently, if $F \in \mathbb{Z}[[q]]$ satisfies $(q; q)_J F \succeq 0$ and $S \subset \mathbb{Z}_{>0}$ is a finite multiset admitting an injection $\sigma: S \rightarrow \{1, \dots, J\}$ with $\sigma(a) \mid a$ for all $a \in S$, then $\prod_{a \in S} (1 - q^a) F \succeq 0$.*

Proof. If $b \mid a$, $\frac{1 - q^a}{1 - q^b} = 1 + q^b + \dots + q^{a-b}$. If $b \nmid a$, the series $\frac{1 - q^a}{1 - q^b} = (1 - q^a) \sum_{t \geq 0} q^{bt}$ has coefficient -1 at the least exponent of the form $a + bt$ not divisible by b (namely a itself if $b \nmid a$; indeed the coefficient of q^n is $\#\{t : bt = n\} - \#\{t : a + bt = n\}$, which is -1 at $n = a$ when $b \nmid a$). For the consequence, write

$$\prod_{a \in S} (1 - q^a) F = \left[\prod_{a \in S} \frac{1 - q^a}{1 - q^{\sigma(a)}} \right] \left[\prod_{b \in \{1 \dots J\} \setminus \sigma(S)} \frac{1}{1 - q^b} \right] (q; q)_J F,$$

a product of three coefficientwise-nonnegative series. $\square$

Empirically (29,482 subsets tested over four (d, c, m, k) cases, zero violations) the divisibility-matching condition also appears close to necessary. However, for the induction step one needs $\{1, \dots, j\} \cup \{m - k - 1\}$ to match into $\{1, \dots, j + 1\}$, which forces $(j + 1) \mid (m - k - 1)$: the Transfer Lemma alone proves only these arithmetic slices, not the full tower.

4 The chain model and two structural lemmas

Cylindric partitions of profile c with largest part $\leq m$ biject (conjugate model, verified against the transfer matrix) with chains $a^{(1)} \geq \dots \geq a^{(m)}$ in $S = \{a \in \mathbb{Z}_{\geq 0}^3 : a_i \leq a_{i-1} + c_i \text{ cyclically}\}$, weighted by $\sum_s |a^{(s)}|$; largest part exactly m iff all levels are nonzero.

Lemma 4 (Slack). *Every $a \in S$ has a slack coordinate i (i.e. $a_i < a_{i-1} + c_i$), since $d = \sum c_i \geq 1$ makes all three constraints tight impossible.*

Lemma 5 (Switch). *If i is tight at a and slack at b , with $a \geq b$ componentwise ($a, b \in S$), then $a_i = a_{i-1} + c_i \geq b_{i-1} + c_i > b_i$, so $a_i > b_i$ strictly.*

These make box-adding ribbon maps well defined, but three distinct designs of a weight- $(+m)$ injection $\psi: C_m \rightarrow C_m$ (needed for $f_0^{(m)} \succeq 0$) failed: a greedy least-slack ribbon is total but not injective; a claim-based inverse-by-construction map is injective but not total (orphan states exist for every profile); a top-level add reduces circularly to $(1-q)g_m \succeq 0$. The pattern of failure suggests a global canonical structure — plausibly a crystal operator on chains (cf. Tingley’s question on lifting $U_q(\widehat{\mathfrak{sl}}_n)$ crystal structures to cylindric plane partitions) — is required.

Key example

$d = 4$, $c = (2, 1, 1)$, $m = 2$: $f_0^{(2)} = (1 - q^2)g_2 - q g_1$ has expansion $3q^3 + 7q^4 + 10q^5 + 13q^6 + \dots \succeq 0$, and $(1 - q)f_0^{(2)} = D_1^2 = 3q^3 + 4q^4 + 3q^5 + 3q^6 + 2q^7 + 2q^8 + q^9 + q^{10} + q^{12} \succeq 0$, while $(1 - q)(1 - q^2)f_0^{(2)}$ has coefficient -1 at q^6 : exact 1-fold monotonicity, matching Master at $(k, m) = (0, 2)$.